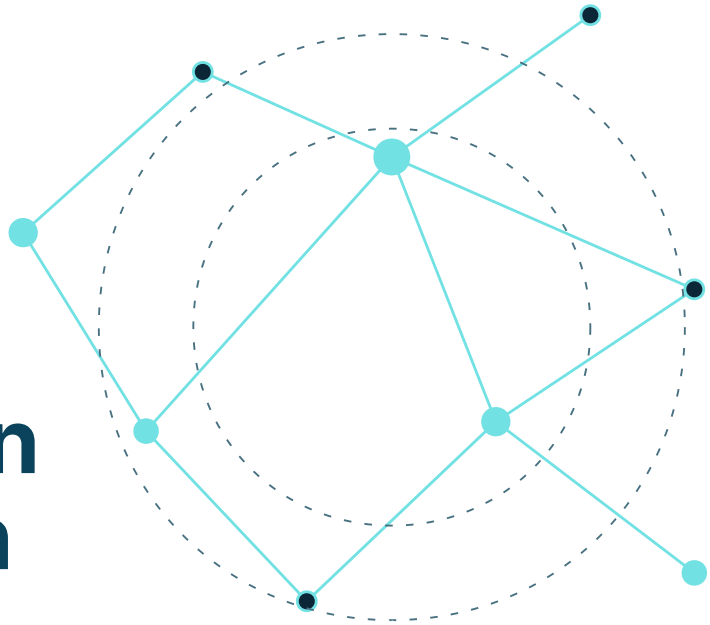
Institute of Aviation Engineering & Technology
Electronics & Communications Engineering Program



PROPOSED HOMEPAGE REDESIGN

Website Redesign & Implementation Specific action

مقترح تطوير الموقع الإلكتروني
لبرنامج هندسة الإلكترونيات والاتصالات

A review-ready design directionand practical frontend hand of for institutional discussion.

PROPOSAL FOR REVIEW
Not an approved institutional publication

Prepared 23 October 2026

Egyptian Aviation Academy
الأكاديمية المصرية لعلوم الطيران

Institute Of Aviation Engineering and Technology
معهد هندسة و تكنولوجيا الطيران

عبدالرحمن مصطفى
ABDELRAHMAN MOSTAFA
SOFTWARE DEVELOPER

Executive overview

The proposal

Abilingual, accessible homepage for IAET's Electronics & Communications Engineering Program, with a clear path from program identity to academic information and official resources.

Why this direction

The conceptorganizes existing institutional content into a deliberate sequence, gives official resources clearer prominence, and introduces a visual language rooted in signal flow and engineering diagrams. These are design goals—not claims of measured problems on the current site.

DECISION SOUGHT

Approve the direction, not yet the final content.

Confirm visual direction, logo/assets, current academic documents, bilingual copy, and permission to proceed to implementation.

Primary audiences

Current students
Find regulations, course specifications, guides and schedules.

Prospective students & families
Understand the program's purpose, mission and academic context.

Staff & visitors
Reachprogram leadership, staff information and official contacts.

Scope

IN SCOPE

One complete program homepage; desktop/mobile direction; bilingual UX; reusable components; design tokens; interaction and implementation guidance.

OUT OF SCOPE

Full institute redesign, backend services, authenticated student portal, new admissions process, verified live AI chatbot, or replacement of official academic documents.

ABDELRAHMAN MOSTAFA
SOFTWARE DEVELOPER

Research & institutional facts

Item		Status / use
Institute name	Institute of Aviation Engineering & Technology (IAET)	Verified on institute pages
Established	1997	Published institutional history
Location	Al Matar, Imbaba, Giza Governorate, Egypt Electronics & Communications Engineering	Published contact location
Program	Electronics & Communications Engineering	Listed as an academic program
Program head	Dr. Mohammed Sobhy	Published director's message; confirm preferred transliteration/title
Contact	info@iaet.edu.eg · 02 33594060	Published on official homepage/contact pages
Accreditation	Program page includes a statement for academic year 2024/2025	Historical/time-bound; do not imply current-year status

Editorial rule. Institute-level accreditation language on the homepage is not a substitute for a current, program-specific statement. Any accreditation claim in the redesigned homepage requires confirmation for the relevant academic year.

Source access & limitations

The English and Arabic program pages and institute overview pages were indexed and readable during research. The published program page includes vision, mission, aims/objectives, leadership message, staff links, program competences, internal regulations, course/program specifications, student guide, surveys and dated study schedules. Some linked documents and underlying media were not independently reviewed in this pass; link availability and currency must be checked before launch.

Sources: iaet.edu.eg/comm · iaet.edu.eg/ar/comm · iaet.edu.eg/ · iaet.edu.eg/ar/home · iaet.edu.eg/aboutus · iaet.edu.eg/ar/aboutus

Official program content

OFFICIAL ENGLISH · VISION

“That the electronic engineering and communications program achieve leadership at the local and regional levels in accordance with comprehensive quality standards in the fields of teaching, learning, scientific research and community services.”

النص العربي المنشور · الرؤية

أن يحقق برنامج الهندسة الإلكترونية والاتصالات الريادة على المستوى المحلي والإقليمي وفق معايير الجودة الشاملة في مجالات التعليم والتعلم والبحث العلمي وخدمة المجتمع.

OFFICIAL ENGLISH · MISSION

“To provide the educational, training and research environment appropriate for students of electronics engineering and communications program. To enable students to keep abreast of rapid developments in technology, competition locally and regionally, and the ability to contribute to community service and the development of the environment within the framework of values and ethics.”

النص العربي المنشور · الرسالة

توفير البيئة التعليمية والتدريبية والبحثية المناسبة لطلاب برنامج هندسة الإلكترونيات والاتصالات. تمكين الطلاب من مواكبة التطورات السريعة في التكنولوجيا والمنافسة محلياً وإقليمياً والقدرة على المساهمة في خدمة المجتمع وتنمية البيئة في إطار القيم والأخلاق.

Published program aims / objectives — editorially normalized

1. Use analysis and simulation tools from engineering science and mathematics.
2. Identify and develop technologies across electronics and communications engineering.
3. Formulate and solve engineering problems through analytical thinking, scientific research and creativity.
4. Work in multidisciplinary environments and communicate effectively.
5. Lead and assume responsibility with professional ethics and respect for intellectual rights.

Content inventory & source policy

Content block	Preferred source	Publication rule
Vision, mission, aims	Program page / Arabic equivalent	Preserve meaning; approval for edits and translation parity
Leadership message	Official program page	Use approved excerpt; confirm current head and portrait rights
Staff and organization	Official staff and chart links	Link to source; verify names, titles and currency
Academic resources	Regulations, competences, course/program specs, student guide	Only publish working links; label academic year/version
Study schedules	Official schedules pages	Existing listed schedules include 2020–2022; archive or replace only after confirmation
Accreditation	Current formal institute/program confirmation	No current claim based only on 2024/2025 statement
Contact and location	Official institute contact pages	Reconfirm before release; use mailto/tel links

Proposed homepage copy — clearly not official

CREATIVE HEADLINE · PROPOSAL

Every signal begins a connection.

من الإشارة تبدأ الحكاية،ومن الهندسة يتصل العالم.

No facilities, equipment, rankings, graduate outcomes, partnerships, testimonials or admissions claims are introduced. The learning-resources section is framed around confirmed links and general academic support rather than invented labs or hardware.

Creative direction

CORE CONCEPT

A signal becomes useful when its path is clear.

Thevisual system treats information asa network:a few deliberate nodes,carefully routedlines, measured spacing and clear labels.The resultistechnicalwithout becoming a telecom-company advertisement.



Two identities, one hierarchy

IAET appears as the institutional parent in the header and footer. The program name remains the page's dominant subject. Use the genuine IAET logo asset only after confirmation; do not invent a program logo.

Emotional intent

Precise, calm, academically serious, welcoming. Strong typographic contrast and a restrained cyan signal accent create memorability without neon effects or excessive gradients.

Visual references — principles, not copied layouts

Kuadra's Arabic site was reviewed for clear content hierarchy, connected narrative, purposeful section rhythm and explicit next steps. Its construction-technology branding, layouts and product aesthetic are not to be copied. The proposed concept uses a distinct academic editorial system.

Homepage information architecture

01 Header → 02 Hero → 03 Program intro → 04 Vision / mission → 05 Objectives

06 Academic resources → 07 Leadership message → 08 Learning resources

09 Student actions → 10 Contact / footer

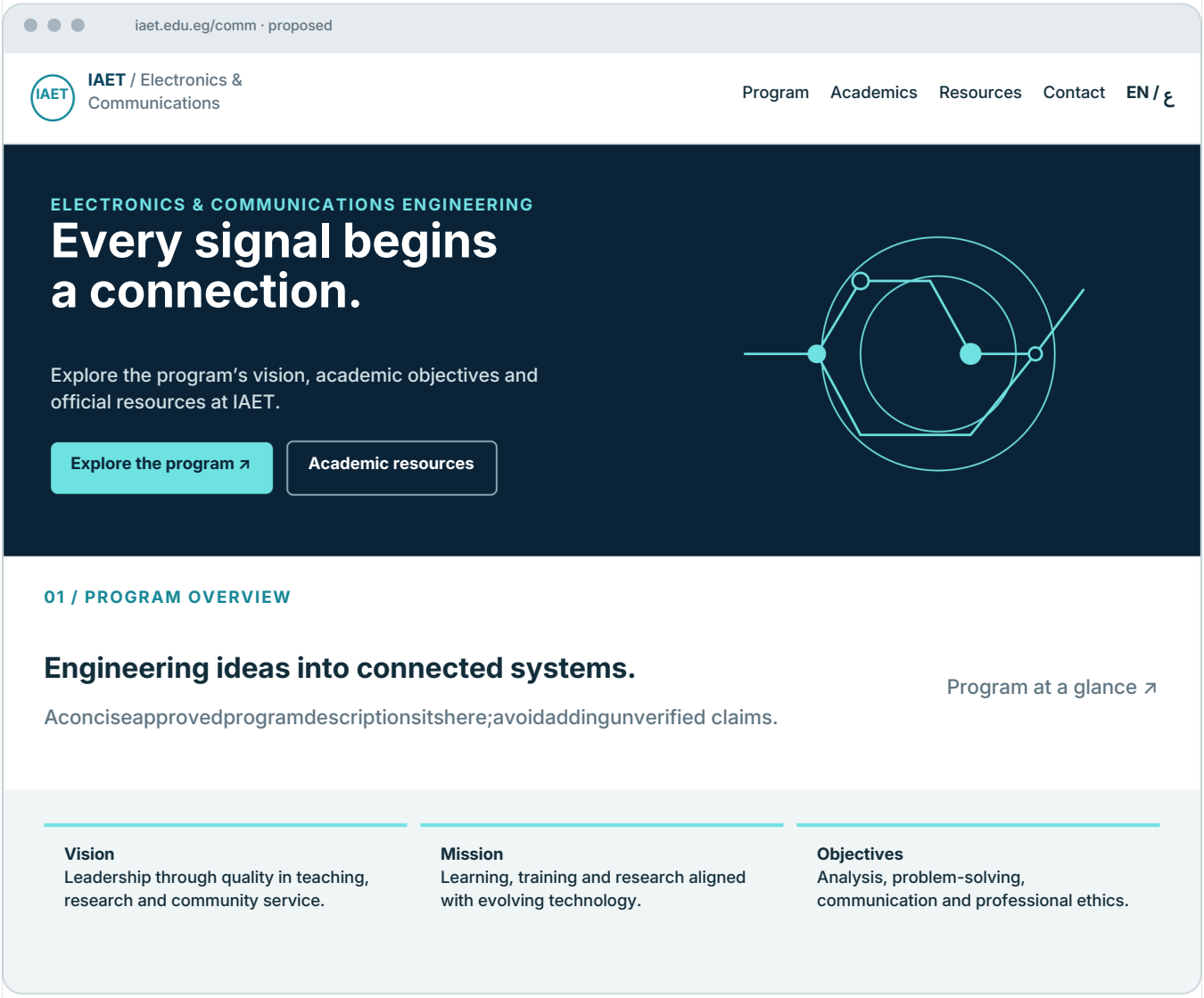
Persistent utility: language switcher and chatbot launcher remain available without obscuring content. Chatbot panel is opened on demand, not auto-opened.

Section	Purpose / audience	Primary action	Factual source
Header + hero	Orient every visitor; establish institution/ program relationship	Explore program / academic resources	Official program and institute names
Program intro	Give context to prospective students and visitors	Read about program	Approved program description (confirm if available)
Vision + mission	Communicate purpose and values	Continue to objectives	Official program page
Objectives	Summarize intended graduate capabilities	Open program specifications	Official aims list
Academic resources	Support current students	Open verified resource	Official linked documents
Leadership	Human institutional context	Read approved message	Program director’s message
Le ar ning resources	Make learning-related information discoverable without unverified claims	Open course specifications / guide	Official academic links
Student CTA + footer	Reduce navigation friction; provide contact	Email / call / visit official pages	Institute contact page

07 / ANNOTATED DESIGN ARTIFACT

Homepage visual — desktop direction

Conceptualinterface composition,notascreenshotofanimplementedwebsite.Thefinal logo asset and approved images are still required.



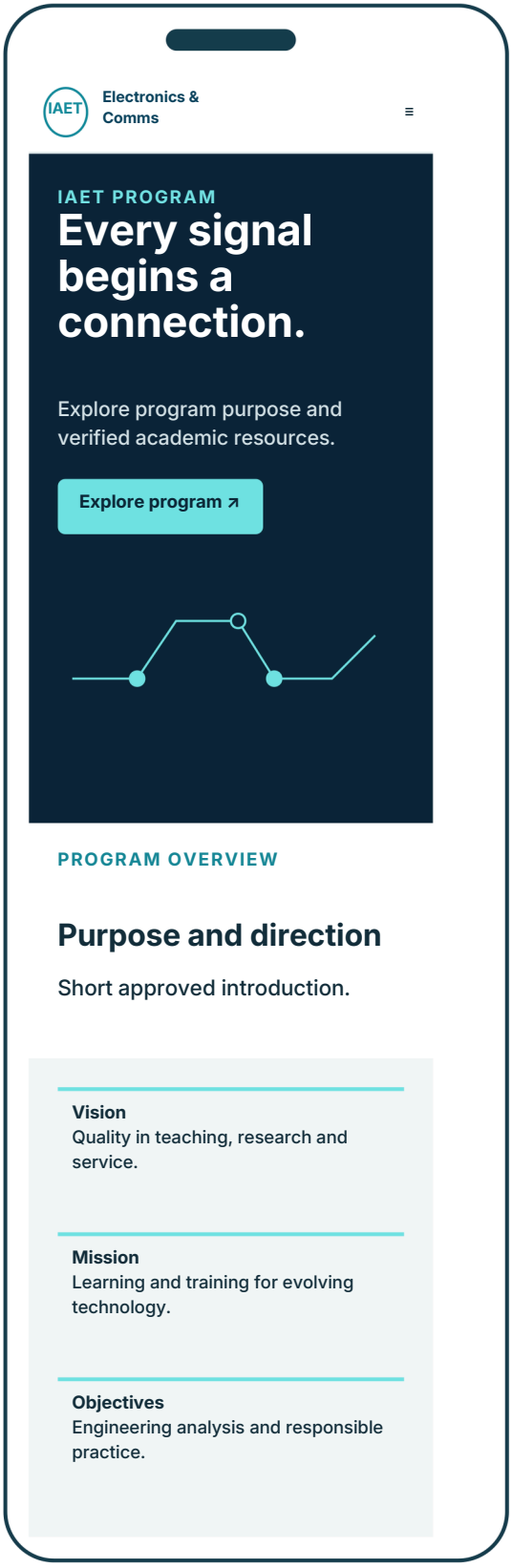
01 Institution/program wordmark lockup; use verified IAET logo in production.

02 Short headline and two clear actions; decorative signal graphic has meaning.

03 Vision, mission and objectives use distinct editorial blocks—not repeated cards.

04 Dark hero anchors the page; warm/light surfaces support reading and printing.

Homepage visual — mobile direction



Mobile rules

• Single-column content; keep heading line lengths short. • Header becomes compact: logo/wordmark, menu button and language control inside menu or compact utility row.

- Primary CTA spans available width when necessary; preserve at least 44 × 44 CSS px touch targets.
- Signal illustration moves below or alongside the headline only when it does not compress copy.
- Objective list and academic resources remain fully readable; no hover-only information.
- Chat launcher sits above the safe-area inset and must not cover primary actions.

Responsive breakpoint proposal: 640px / 768px / 1024px / 1280px. Treat these as starting tokens; test actual content wrapping in both directions.

RTL adaptation

For Arabic,align textand navigation to the right, use logical CSS properties, mirror directional arrows where meaning changes, and keep the signal motif direction-neutral. Do not reverse brand marks or technical diagrams automatically.

Component specification

Component	Content / desktop	Mobile + states	Accessibility / RTL
Site Header	Verified IAET logo, program label, nav links, language switch	Collapses to menu; closed/open states; Escape closes	Semantic header/nav, focus trap only if modal; logical alignment
Hero	Eyebrow, proposed headline, short approved description, two CTAs, signal SVG	Stacked; art moves below text as needed	Decorative SVG aria-hidden; heading is page h1
ProgramIntro	Short approved overview and contextual link	One column, no tiny side copy	Use headings in order; no unsupported program claims
VisionMission	Two editorial panels with official statements	Stacked blocks, clear Arabic direction	Language metadata per localized page
ObjectivesList	Numbered list or staggered typographic rows	Plain numbered list	Native ol; do not encode numbers in images
AcademicResources	Grouped links: program, course specs, student guide, regulations, staff	Full-width rows with title and description	Descriptive link names; open same tab unless file/download behavior is clear
LeadershipFeature	Approved excerpt and verified portrait only if available	Text first; portrait optional	Alt text for meaningful portrait; quote attribution
Le ar ningResources	Official academic destinations; no invented lab gallery	List or two-column at wide screens	Only verified current links; date labels on documents
StudentActions	Clear resource CTA and contact path	Stacked buttons	Visible focus and accessible names
SiteFooter	Institute name, location, email, phone, official links	Stacked columns	mailto/tel; Arabic/English values consistent
ProgramChatbot	Launcher + panel, suggested questions, cited source links	Near-full-screen panel on small phones	Keyboard reachable; announces messages; no forced focus on launch

Interaction states to design: default, hover, keyboard focus, active, expanded/collapsed, loading (chat only), empty/unknown response, and link unavailable.

Design system & tokens

Color tokens

Token	HEX	Use
Institutional navy	#0B435D	Primary headings / institutional anchor
Deep ink	#0B2737	Hero / dark surfaces
Signal cyan	#72E1E3	Signal accents, not small text on white
Cloud	#F2F6F7	Secondary surface
Warm white	#FFFFFF	Primary canvas
Body ink	#122D3A	Body text
Muted	#607582	Secondary text (check contrast by size)

Typography

Arabic: NotoSans Arabic — clear shaping and broad coverage. **English:** Inter — neutral, readable technical editorial voice.

Token	Size / line-height
Display	48–64 / 1.05 40–
H1	48 / 1.1
H2	28–36 / 1.2 20–
H3	24 / 1.25 16–18 /
Body	1.6
Caption	12–14 / 1.45

Scale in CSS px; PDF uses an independent print scale. Arabic line-height should generally be 1.7–1.9 where diacritics and ascenders need room.

Spacing

4,8,12,16,24, 32, 48, 64, 96 px

Layout

Maxcontent 1200px; 12-col desktop grid; 24px gutters; 16px mobile gutters.

Shape

2–4px radius for controls; 6px for panels; minimal shadows; 1px neutral borders.

Contrast: target WCAG 2.2 AA: 4.5:1 for normal text, 3:1 for large text and meaningful UI boundaries. Cyan is an accent on dark backgrounds, not a default text color on white. Verify exact token pairs in implementation.

All values are proposed design tokens, not confirmed IAET brand standards. Obtain brand guidelines and official logo asset before final implementation.

Motion & interaction

Element	Trigger	Motion	Fallback
Hero text	Page load	Opacity + 12px vertical shift, 450–600ms, ease-out, stagger ≤100ms	Immediate render with reduced motion
Signal graphic	Page load / subtle scroll	Draw/reveal line or 4–8px parallax, 600–900ms; never continuously pulse	Static SVG
Section content	Enters viewport	Opacity + 10px translate, 350–500ms, once only	No animation under prefers-reduced-motion
Buttons / links	Hover, focus, press	Color/border transition 120–180ms; no scale bounce	Strong focus outline
Mobile menu	Tap menu / Esc ape	Short slide/fade 180–240ms	Works with motion disabled
Chat panel	Launcher activation	Fade/translate 180–240ms	Panel opens without animation

Technology responsibilities

Framer Motion
Component-level entrance states, menu/panel transitions and reduced-motion variants.

GSAP
Use only if a specific signal-path drawing or coordinated scroll sequence genuinely needs timeline control. Avoid duplicating the same animation in both libraries.

> **Performance rule:** animate transform and opacity; avoid scroll-jacking, long pinned sections, large blur filters, continuous motion, and animation that delays access to content. Respect `prefers-reduced-motion`.

Interaction outcomes

- Language switchchangesroute/content and document direction, not just alignment.
- Mobile menu reports expanded state and closes after navigation.
- Resource links show a clear label and open the correct official destination.
- Chatbot unknown answers state the limitation and suggest official links.
- All actions remain usable with keyboard and touch; hover is never the only signal.

Arabic / English implementation

ENGLISH / LTR

Electronics & Communications

Navigationand text align left; inlineicons follow reading direction where appropriate. English page uses lang="en" dir="ltr".

Program → Academics →

Resources

العربية / RTL

هندسة الإلكترونيات والاتصالات

تبدأالمحاذاة ومساراتالتنقل مناليمين. استخدم خصائص CSS المنطقية،ولاا تعكسالرسوم التقنية أو الشعارات تلقائياً.

البرنامج ← الدراسة ← المصادر

صفحة العربية تستخدم "lang="ar" dir="rtl .

Implementation rules

- Use localized routes, e.g./en/comm and /ar/comm, or retain current official routes if URL compatibility is required; confirm routing with IAET before release.
- Store translations in structured content data, not scattered inline ternaries. Keep the same semantic content and equivalent resource destinations in both languages.
- Set lang and dir on the document root for each locale; use dir="auto" only for user-generated or mixed-direction snippets where needed.
- Prefer CSS logical properties: margin-inline-start, padding-inline, inset-inline-end. Avoid hard-coded left/right layout assumptions.
- Mirror directional chevrons and back/forward arrows when meaning depends on direction; do not mirror play, external-link, signal-flow or technical symbols automatically.
- Arabic body text: allow longer line height and slightly wider text measures; verify punctuation, numerals, phone number display and email rendering.
- Language switch preserves the current page/context where an equivalent route exists; otherwise send the visitor to the localized homepage and announce the change.

قاعدة أساسية: الترجمة المعتمدة ومراجعة المحتوى العربي مسؤولية البرنامج قبل النشر؛ لا يكفي قلب اتجاه الصفحة لإنتاج تجربة ثنائية اللغة سليمة.

عبدالرحمن مصطفى
ABDELRAHMAN MOSTAFA
SOFTWARE DEVELOPER

Next.js frontend handoff

Suggested App Router structure

```
src/
  app/
    [locale]/comm/page.tsx
    [locale]/layout.tsx
    globals.css
  components/
    site-header.tsx mobile-
    navigation.tsx hero-
    section.tsx program-
    intro.tsx vision-
    mission.tsx objectives-
    list.tsx academic-
    resources.tsx
    leadership-feature.tsx
    site-footer.tsx
    program-chatbot.tsx
  content/
    en/comm.
    ts
  lib/r/comm.t
    focale.ts
    metadata.ts
  public/
    brand/ (verified logo assets only)
    images/ (approved photos)
    diagrams/ (authored SVGs)
```

Content model

Typed content objectsfor navigation, hero, vision, mission, objectives, resource groups, leadership excerpt, contact and chatbot FAQ. Resource entries should include label, localized label, URL, resource type, source owner and last-verified date.

SEO & metadata

Localized title/description, canonical URLs, hreflang alternates, Open Graph metadata, semantic headings and meaningful internal links. Do not add unsupported

accreditation schemaor fabricated organizationdata.

Rendering & assets

Preferserver-rendered static content; isolate menu, language controls and chatbot into client components. Use optimized images and SVG for authored signal artwork. Avoid loading animation libraries on sections that do not use them.

Reliability

Validate resource URLs, handle missing images, test route parity, avoid nested interactive elements, and make unknown localization keys fail visibly during development.

Suggested implementation sequence

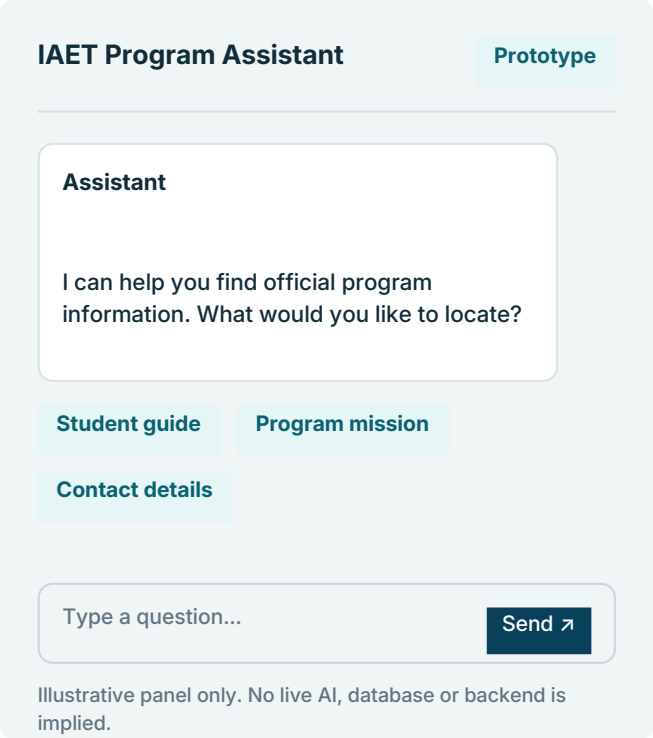
Buildin this order: design tokensandapproved content;header and hero; content sections and resources; locale routing and RTL; local FAQ chatbot; then full QA and review.

Architecture isarecommendation, notarequirementtorestructuretheinstitute'sexistingapplication.Checkthecurrent deployment, route conventions and dependency versions before implementation.

Program-information chatbot

Role and placement

Acompact helplayer for discovering verified program information and official academic resources. A fixed launcher sits in a lower corner, respects mobile safe areas
and never opens
automatically.



Illustrative panel only. No live AI, database or backend is implied.

Suggested starter prompts

- Where can I findthe studentguide?
- Where are the course specifications?
- What is the program's mission?
- How do I contact the institute?
- Where can I find staff information?

Response and safety rules

- Prototype v1 usesan approved local FAQ/content map; answers only from reviewed program facts and resource URLs.
- Every factual answer includes a "Source" link to the official page/document used.
- Unknown or ambiguous questions receive a clear message: "I can't verify that from the program information available here," followed by relevant official contact/resource links.
- Never invent current timetables, admissions requirements, accreditation status, staff roles or facilities. •

Arabic interface uses Arabic starter prompts and equivalent approved content; direction and message alignment follow locale.

- Keyboard: launcher has an accessible name; panel has a title, close button and logical tab order; Escape closes; messages are announced via a polite live region.
- Mobile: panel becomes near-full-screen with scrollable transcript and persistent composer; do not cover browser controls or essential page CTAs.

> **Future extension:** a live AI service would require institutional approval, a maintained source corpus, privacy review, monitoring and a defined escalation path. None is assumed in this proposal.

Responsive & accessibility QA

All items belowareproposedtests.Theyhavenotbeenmarkedaspassed because the website has not yet been implemented and tested.

Viewport matrix

- ☐ 320–375px small mobile
- ☐ 390–430px larger mobile
- ☐ 768px tablet portrait
- ☐ 1024px tablet / small laptop
- ☐ 1280px laptop
- ☐ 1440px+ large desktop
- ☐ Zoom at 200% without loss of content

Language & content

- ☐ Arabic RTL at every viewport
- ☐ English LTR at every viewport
- ☐ Equivalent navigation and resources
- ☐ Long Arabic headings / mixed numerals
- ☐ Current links and document labels
- ☐ Phone/email copy and tap behavior

Accessibility

- ☐ Keyboard-only navigation
- ☐ Visible focus on all controls
- ☐ Logical heading order and landmarks
- ☐ Contrast meets WCAG AA
- ☐ Reduced-motion preference
- ☐ Screen-reader names and announcements
- ☐ Touch targets at least 44 × 44 CSS px
- ☐ Menu/chat can open and close by keyboard

Functional / performance

- ☐ No broken links or missing assets
- ☐ Language switch preserves context
- ☐ Chatbot unknown-answer behavior
- ☐ No console errors or hydration mismatch
- ☐ Responsive images and optimized fonts
- ☐ Core Web Vitals reviewed in staging
- ☐ Print and narrow-screen layout checked

Release gate: do not publish academic resources or accreditation wording until the program confirms that each item is current and authorized. Record test date, browser/device and outcome in the implementation issue tracker.

Approval & implementation roadmap

01 · Review creative direction — program head confirms overall visual approach and homepage hierarchy.

02 · Verify institutional assets and content — obtain official logo files, approved Arabic/English copy, current resource links and leadership details.

03 · Implement homepage — build the approved React / Next.js page and reusable components.

04 · Test bilingual experience — validate RTL/LTR, keyboard access, responsive behavior, links and performance.

05 · Demonstrate to leadership — review working browser version with program leadership.

06 · Apply feedback — record decisions, resolve content issues and retest.

07 · Consider expansion — only after approval, evaluate reuse for other IAET departments.

Decisions requested from the program head

01 — Design direction
Approve orrevise the signal/network concept, proposed palette and content order.

02 — Official identity
Provide/confirmthegenuine IAET logo, brand guidance and authorized imagery.

03 — Content authority
Confirmcurrent vision/mission, objectives, director's message, staff and academic links.

04 — Proceed to build
Authorize the homepage implementation and nominate the reviewer for bilingual content.

CLOSING NOTE

A clearer digital front door for the program.

This proposal is a review document and implementation reference. It is not an official IAET publication, does not imply institutional endorsement, and should not be treated as final until content and identity decisions are approved.

ABDELRAHMAN MOSTAFA
SOFTWARE DEVELOPER

References & factual notes

Official IAET sources reviewed

Source	URL	Used for
Program page English	https://www.iaet.edu.eg/comm	Vision, mission, aims, leadership message, resource link inventory
Program page Arabic	https://iaet.edu.eg/ar/comm	Arabic statements, staff/resource link labels
Institute home page	https://iaet.edu.eg/	Institute name, founding year, program list, contact details
Arabic homepage	https://iaet.edu.eg/ar/home	Arabic institutional overview and contact
About IAET — English	https://iaet.edu.eg/aboutus	History, institute vision and mission
About IAET — Arabic	https://iaet.edu.eg/ar/aboutus	Arabic institutional history and statements
Kuadra — Arabic	https://www.kuadra-eg.com/ar	Benchmark for hierarchy, narrative rhythm and explicit actions only

Factual and editorial notes

- Verified from official indexed pages: IAET states it was established in 1997; the institute homepage lists Al Matar, Imbaba, Giza, info@iaet.edu.eg and 02 33594060.
- Program page publishes a vision, mission, aims, director's message and academic-resource links. The English source's objective-list introduction appears malformed; objectives in this proposal are editorially normalized and require approval.
- The program page includes an accreditation statement for 2024/2025. This proposal deliberately does not treat it as proof of status in 2025/2026 or 2026/2027.
- Official logo asset and current brand guidelines were not verified/downloaded for inclusion. The cover's circular IAET mark is a clearly conceptual placeholder, not a redraw or official logo.
- Proposed colors, fonts, signal graphics, headline and layout are design recommendations, not official identity rules.
- Linked PDF/document resources, image rights, current leadership details and the current accreditation position remain to be confirmed by IAET.

Research boundary: official webpages were reviewed via accessible indexed page content on 23 October 2026. This is not a legal, accreditation or comprehensive document audit; source pages and links should be rechecked at implementation time.

Prepared for institutional review · Version 1.0 · 23 October 2026

ABDELRAHMAN MOSTAFA
SOFTWARE DEVELOPER